PCC2 Hand Group Peer Review 1:

- There is repeated use of very informal language. It feels more like it was written as a script of a conversation than as a lab report. Could be improved.
- There is much use of first-person language which is fine (mainly 'we'), but sometimes it crosses the line into being too informal. Should reconsider how you want the report to be perceived by the reader.
- There are many different acronyms used. It may be helpful to have a glossary somewhere for easy referencing.
- There doesn't seem to be any real mention of what has been done so far asides from some research, exercise development and testing. It mentions that current results demonstrate feasibility, but what results are these? Has the device itself even been assembled or tested? This is not stated.
- How do you plan to test this, particularly in relation to the effectiveness of the electrical design and the developed exercises. There doesn't seem to be any testing so far or any plans to do so.
- The overall structure could be improved to make it more in line with general report strucure. A dedicated background section would help with repetition and consolidate relevant information. Having a methods and results section (with subsections if necessary) would clarify the differences between research done, work done and results achieved.
- Lots of short/one sentence sections that should be combined or removed.
- In Chapter 3 particularly there is a lot of repetition between the 'Measured Signals' and 'Extracted Metrics' sections. Some of these don't make sense, and overall should probably just be stated once. Could maybe just remove all instances of 'Measured Signals'.
- There are several sections that provide a lot of information in a list with little separation. The format of these could be improved/simplified.
- The lack of clear methods means that reproduction is impossible.

Smart Hand Rehabilitation System

Naya Nasr Lucia Saez Perez Iaroslav Petrishchev

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1 Introduction

Neurological and neuromuscular disorders such as stroke, traumatic brain injury, and Parkinson's disease frequently lead to tremor. Tremors are involuntary muscle movements that produce a visible shaking in one or more body part [3]. Other common symptoms include muscle weakness, reduced motor control, fatigue, bradykinesia, and abnormal co-contraction. Because hand function plays a critical role in activities of daily living, its deterioration significantly impacts patient independence and quality of life. Patients who suffer from tremor usually go through occupational therapy, where they are asked to perform repetitive tasks, high intensity training and fitness training to potentially imporve motor recovery of the arm. [7].

Traditional clinical tools for hand assessment, such as dynamometers, only measure maximum voluntary contraction (MVC). While MVC is an important parameter, it is not enough to fully understand if a patient is improving from the rehabilitation therapy or not.

Studies have shown that the use of several sensors to assess the severity of tremor can be usefull. Ali, F. et al. [2] suggests the usage of IMU sensors while the patients is asked to perform tasks. Other studies suggest the implementation of EMG sensors to provide an insight into neuromuscular activation patterns and fatigue.

The objective of this project is to design and develop a **Smart Hand Rehabilitation System** that integrates pressure (insted of dynanometer due to cost limitations), EMG, and IMU sensor, into a portable, Bluetooth-enabled device capable of real-time monitoring and feedback. The system is intended to assess two primary clinical populations:

- Stroke or brain injury patients.
- Parkinson's or tremor-dominant patients.

Rather than focusing solely on peak strength, the proposed system extracts clinically meaningful biomarkers including maximum force, rate of force development, fatigue index, force variability, tremor band power, tremor frequency, orientation stability and many more that will be explored later in the report.
By analyzing both mechanical output and neuromuscular activation, the system enables deeper interpretation of motor performance, to therefore understand if a patient is improving or deterioration after rehabilitation therapy.

1.1 Members' contributions

Naya Nasr - Programming of the sensors on NRF52, Implementation and testing of data acquisition from the sensors, for tremor measurements. Research on the feature extraction methods, with a focus on the pressure sensor. Development of the current web application for real-time data visualization.
Lucía Pérez Saéz – Organized target patient groups and designed exercise protocols based on clinical evidence. Planned the signal processing for each sensor feature and researched the scientific evidence supporting their selection, focusing on IMU features. Conceptualized the Phase 1 rule-based algorithm, including workflow, thresholds, and logic for real-time feedback.
Iaroslav Petrishchev - Design of the general system architecture. Design of the device's electrical circuit, preparation of schematics and connections. Search and procurement of required components, as well as assembly of the device prototype and initial testing of individual components.

2 System Overview and Hardware Design

The overall plan of the system implemented in this project is presented in the block diagram in Figure 1.

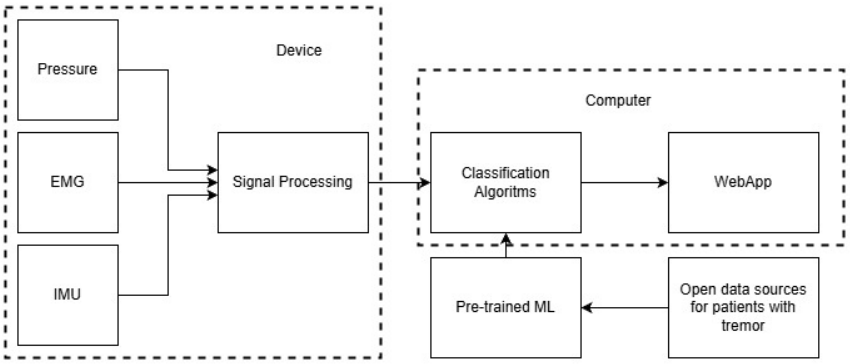


Figure 1: System diagram

It is worth explaining which parts the system consists of. In a general sense, we can divide it into two parts. The **Device** part, which is responsible for aggregating all necessary data, as well as the **Computer** part, which is responsible for processing the received data and presenting it to the user.

2.0.1 Device

The device consists of the main NRF52840 board and three sensors: a pressure sensor (ABPDRRV060MGAA5), IMU sensors (a set of triaxial accelerometers and gyroscopes inside the NRF52 board), as well as an EMG sensor (Muscle Sensor V3.0). More details about each of the sensors and circuit components will be given in the next chapter of the report. In addition, the device includes battery systems to supply power to all elements. The detailed connection of all components is shown in Figure 2.

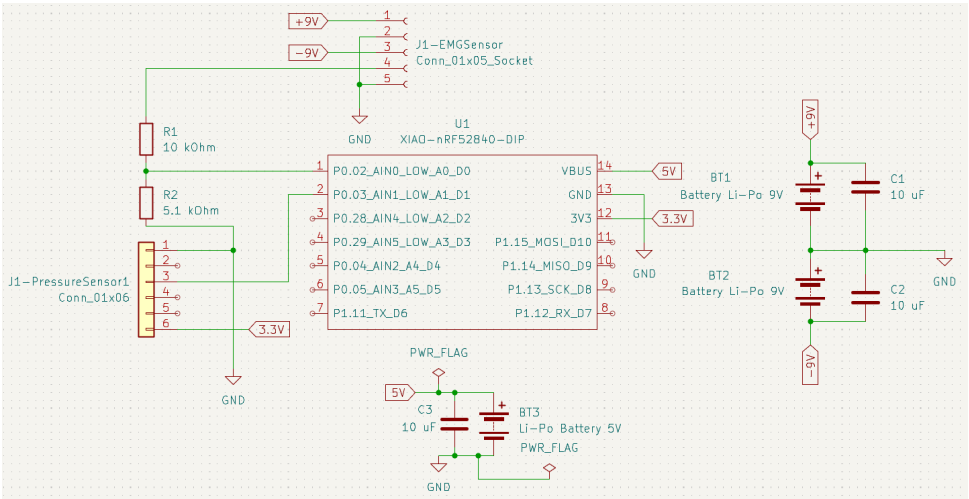


Figure 2: Enter Caption

It is worth noting that the IMU sensors do not require additional wiring, since they are built into the board, and they can be accessed programmatically directly. The pressure sensor is analog, consumes 3.3 V from the board, and outputs an analog signal to pin D1. The EMG sensor is connected in a more complex way. A discussion of its choice and usage specifics will be in the next chapter. For

now, we will say that this sensor is, in fact, an amplifier of the potential difference that we obtain on the electrodes, and therefore a bipolar 9 V supply is required. After that, the analog signal we obtain is passed to pin D0 through a voltage divider to protect the board, since the peak values of the sensor can theoretically reach 9 V, but the board pins can accept only 3.3 V. For the voltage divider we use 10 k Ω and 5.1 kΩ resistors. To power this sensor, a source different from the board's source is required, since we are not able to output 9 V from the board. For this, two 9 V LiPo batteries are used to create a virtual ground and a bipolar supply. Also, for stabilization and smoothing of the voltage, we use two 10 µF capacitors each. Also, to support the operation of the board itself, we need a 5 V LiPo battery, whose voltage we supply to a special charging pin on the board. To stabilize the voltage on it, we also use a 10 µF capacitor.

2.0.2 Computer

The second important part of the system is the computer, which receives the data collected by the board via Bluetooth. On the computer, the received data is processed and a certain conclusion is formed about certain parameters that symbolize the patient's health state. At the first stage, testing of standard threshold-based classification algorithms is assumed, taking into account a basic set of the most significant parameters. At the second stage, training of a machine learning model on publicly available patient data is assumed, and then using the pretrained model for classification and assessment of the patient's condition. The algorithmic component is described in more detail in separate chapters of the report.

After data analysis, the results are presented in a dedicated application on the computer, which in a clear and accessible form allows a specialist to review the testing results for further conclusion.

2.1 Sensors and Components

Now it is worth talking about the sensors and components used in the project and their specific features. All components used in the project are listed in Table 1.

Component	Model / Type	Role / Notes
Main board	NRF52840 board (NRF52 family)	Core device controller; data aggregation; Bluetooth communication to PC
IMU sensors	On-board IMU (3-axis accelerometer + 3-axis gyroscope)	No additional wiring required; accessed programmatically directly from the board
Pressure sensor	ABPDRRV060MGAA5	Analog sensor; powered from 3.3 V; analog output connected to pin D1
EMG sensor module	Muscle Sensor V3.0	Differential biopotential amplifier; requires bipolar ±9 V supply; analog output to ADC through voltage divider
Voltage divider resistors	10 kΩ and 5.1 kΩ	Used to protect the board ADC input (limit the EMG output level to within 3.3 V range)
Supply smoothing capacitors (EMG)	$2 \times 10\,\mu$F	Stabilization/smoothing for the bipolar EMG supply
Supply stabilization capacitor (board)	$10\,\mu$F	Stabilization/smoothing for the board supply line
Batteries for EMG supply	2×9 V LiPo	Used to create a virtual ground and provide bipolar supply for the EMG module
Battery for board supply	5 V LiPo	Powers the main board via a dedicated charging/power pin

Table 1: Main components used in the project

In essence, the IMU and pressure sensors do not present significant complexity, since the principles of their operation and connection are quite stable and simple. Their use does not cause problems. The EMG sensor, however, is much more controversial and problematic.

2.1.1 Pressure Sensor

Regarding the pressure sensor, it will be connected through a plastic hose to the rubber bulb that the patient will be asked to squeeze to perform some of the exercises. This can be useful to understand the MVC and other Biomarkers that will be explained in details later. Initially we wanted to use a dynanometer, but because of cost limitations, we could not use one that would measure the values of the squeeze in real time to study tremor. The sensor module itself will be located on the device board , which will be attached to the patient's arm , and the tube itself will go from the one to the rubber bulb in the patient's palm.

2.1.2 EMG Sensor

First of all, let us explain what the EMG sensor measures. It measures the potential difference between two electrodes placed on a specific muscle. This potential difference at rest is in the microvolt range, and during active muscle contraction can reach units of millivolts, which is quite small, and this is why the sensor uses an amplifier. At the same time, an important component of this system is the third electrode, which is placed on a muscle not involved in the movement, or on an area without muscles (elbow or wrist). This is necessary in order to create a ground reference for controlling the common potential and for reducing noise in the signal. An example of electrode placement can be seen in Figure 3 below.

As can be seen, in our case the wrist is chosen as the ground, and the measurements will be performed on the muscle (flexor digitorum superficialis), since it is the main finger flexor and is most actively involved in the process of squeezing the pressure-sensor ball, which allows us to better measure patient parameters. It is important to note that the electrodes must be placed along the muscle fibers rather than across them in order to measure the potential difference, and they should be positioned at a distance of no more than 2 cm; otherwise, it will be impossible to measure anything. In addition, it is important to use good gel electrodes that fit tightly on the skin and minimize the impedance between the body and the metallic part of the electrode.

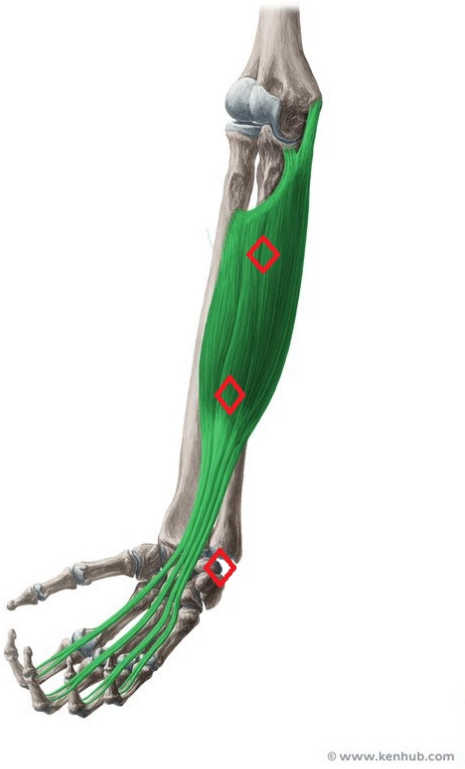


Figure 3: Electrode placement locations for the EMG sensor

Now it is worth mentioning the nuances of such measurements. The EMG sensor is extremely sensitive to mains noise and, because of this, strongly depends on the length and noisiness of the connection wires. The electrodes themselves are equipped with the necessary connectors for noise suppression; nevertheless, the outputs and inputs of the sensor module remain the weak point in this chain, which makes it impossible to properly test the sensor using an oscilloscope and external power supplies. The solution to this problem is to reduce the length of the chains as much as possible, including the use of batteries for power supply and the board for direct signal analysis. Otherwise, the mains noise becomes so strong that the EMG signal becomes indistinguishable even under digital filtering conditions. One solution to this problem is the use of ready-made EMG systems, but they are quite expensive. As an alternative, it is also possible to use the reference-less EMG sensors provided

to our group by the laboratory. They use a virtual ground and perform measurements directly on the electrodes (which are part of the board) and do not require bipolar 9 V power, which makes them less dependent on additional power routing through cables. Their use appears more stable and reasonable, and our group will conduct a number of additional tests and experiments to determine the more preferable EMG system.

2.1.3 IMU Sensor

The Inertial Measurement Unit (IMU) measures linear acceleration and angular velocity in three axes. By integrating these signals, it is possible to estimate hand orientation, movement patterns, and tremor characteristics. In our system, the IMU is integrated directly on the NRF52 board.

For accurate measurements, careful positioning on the hand is required. Ideally, the IMU should be fixed on the dorsal side of the hand, close to its center of mass, to capture both voluntary movements and involuntary tremor. The orientation of the sensor should be consistent across trials to ensure comparable data. For initial prototyping, the IMU can be embedded in a glove that the patient wears while performing the exercises, but further testing is needed to determine the optimal placement for signal accuracy and patient comfort.

3 Experimental Protocols

The Smart Hand Rehabilitation System is designed to support patients during upper-limb rehabilitation by providing real-time quantitative feedback on strength, motor control, and neuromuscular performance. Unlike conventional dynamometers, the system evaluates not only maximal force but also temporal dynamics, fatigue, repeatability, and movement quality. This enables continuous monitoring without requiring the constant presence of a clinician.

To achieve this functionality, two target patient groups have been identified. For each group, specific exercises are designed to extract relevant features that allow meaningful assessment and personalized feedback.

3.1 Target Groups

The proposed system is tailored for two primary patient populations:

- **Stroke / Brain Injury Patients:** These patients typically exhibit reduced strength, delayed muscle activation, impaired coordination, and decreased rate of force development (RFD).
- **Parkinson's / Tremor-Dominant Patients:** These patients are characterized by tremor, force instability, and impaired fine motor control, rather than purely reduced strength.

Each group requires distinct evaluation protocols to capture the relevant physiological impairments and to guide appropriate rehabilitation strategies.

3.2 Stroke / Brain Injury Exercises

3.2.1 Exercise 1: Maximal Grip Strength and RFD

Protocol: The patient is instructed to squeeze the instrumented object as hard as possible for 3–5 seconds. This is repeated 3–5 times with sufficient rest between trials to avoid fatigue.

Measured Signals:

- Force (pressure sensor)
- Muscle activation (EMG)

Extracted Metrics:

- **Pressure:** Peak force (MVC), Rate of Force Development (RFD), Time to peak force
- **EMG:** Peak activation (RMS), Electromechanical delay (EMD)
- **IMU:** Movement artifacts, compensatory motion

Justification:
Maximal grip strength is a well-established and clinically relevant indicator of upper-limb function in post-stroke rehabilitation. It has been shown to be a reliable measure of recovery and a useful prognostic indicator of functional outcomes [11].

However, maximal force alone does not fully capture neuromuscular performance. Previous studies have demonstrated that grip strength assessment should also include temporal and endurance-related characteristics of force production [8]. In particular, stroke patients exhibit impairments not only in peak force but also in the ability to sustain force over time.

Sustained grip tasks have been used to evaluate fatigability, showing that stroke patients experience a faster decline in force during prolonged contractions. For example, [8] quantified how quickly force drops below a given percentage of maximal voluntary contraction (e.g., 80% MVC), demonstrating increased fatigability during the early stages of recovery.

These findings support the inclusion of both maximal grip force and temporal force characteristics, such as the rate of force development (RFD), to better capture neuromuscular deficits after stroke.

Furthermore, combining force measurements with EMG enables the estimation of electromechanical delay (EMD), providing additional insight into neuromuscular activation and coordination, which are commonly affected in neurological populations.

Therefore, this exercise is designed to capture strength, temporal dynamics, and neuromuscular efficiency, enabling a more comprehensive assessment of motor recovery.

While Exercise 1 focuses on maximal force production and rapid activation, it does not capture the patient's ability to regulate and sustain force over time. Therefore, a complementary submaximal task is introduced.

3.2.2 Exercise 2: Target Hold (30–50% MVC)

Protocol: The patient is instructed to squeeze and hold the object at a target force level of 30–50% of their personal MVC (measured in Exercise 1) for 10–30 seconds per trial. The task is repeated 3–5 times with rest intervals.

Measured Signals:

- Force (pressure sensor)
- IMU (optional, for movement stability)

Extracted Metrics:

- **Pressure:** Force variability (SD/CV), Fatigue Index ($\frac{Peak-Final}{Peak}$), Time in target band (
- **EMG:** Median frequency shift (fatigue), co-activation index, RMS
- **IMU:** Orientation stability, smoothness during hold

Justification:
While maximal contractions quantify strength capacity, submaximal sustained grip tasks provide critical insight into motor control, force regulation, and endurance. These aspects are frequently impaired after stroke due to altered motor unit recruitment and increased co-contraction.

A systematic review by *Latorre et al.* highlights that grip force–time curve analysis commonly employs contractions at 20–50% of MVC to evaluate force steadiness and fatigue. The review reports that individuals with neurological impairments exhibit increased force variability and reduced ability to maintain stable force output over time.

This supports the use of controlled submaximal holds in the proposed protocol, enabling the quantification of:

- **Motor control:** through force variability and time within the target band
- **Endurance:** through sustained force maintenance
- **Fatigue:** through decline in force and EMG spectral changes

Therefore, this exercise complements maximal force assessment by capturing dynamic aspects of neuromuscular performance that are essential for functional hand use.

3.3 Parkinson's / Tremor Exercises

3.3.1 Exercise 1: Steady Hold and Tremor Control

Protocol: The patient is instructed to hold the instrumented object at a constant, comfortable grip force (e.g., 20–30% of MVC or a fixed low force threshold) for 15–30 seconds while minimizing involuntary oscillations.

Measured Signals:

- Force (pressure sensor)

- IMU (acceleration and angular velocity)

Extracted Metrics:

- **Pressure:**Tremor amplitude (force oscillations), force variability (SD/CV), spectral power in tremor frequency band
- **EMG:** Burst activity, frequency content, co-contraction patterns
- **IMU:** Tremor amplitude, dominant frequency, orientation stability

Justification:
s Tremor-dominant conditions such as Parkinson's disease are primarily characterized by rhythmic, involuntary oscillations rather than pure strength deficits. These oscillations typically occur within specific frequency bands and can be quantified through both force output and limb motion.

Previous work has shown that pathological tremor can be reliably characterized through frequency-domain analysis, with Parkinsonian tremor typically appearing in the 4–6 Hz range [4]. Measuring fluctuations in grip force and hand motion therefore provides a direct and non-invasive method to assess neuromuscular instability.

Furthermore, increased force variability and irregular motor unit activation are key indicators of impaired motor control in these patients. By requiring the patient to maintain a steady force output, this task isolates the ability to suppress tremor and stabilize motor performance.

This justifies the inclusion of:

- **Force variability and oscillations** → quantify instability
- **Frequency analysis** → identify tremor characteristics
- **EMG patterns** → assess abnormal muscle activation and co-contraction

While the steady hold task isolates the patient's ability to suppress involuntary oscillations, it does not capture deficits in rhythmic motor execution. Therefore, a dynamic grip-release task is introduced.

3.3.2 Exercise 2: Rhythmic Grip-Release Task

Protocol: The patient is instructed to repeatedly squeeze and release the object at a predefined rhythm (e.g., guided by a metronome at 0.5–2 Hz) for 20–40 seconds.

Measured Signals:

- Force (pressure sensor)
- EMG
- IMU

Extracted Metrics:

- **Pressure:** Cycle timing, peak force consistency, inter-cycle variability
- **EMG:** Activation timing, coordination between flexors and extensors
- **IMU:** Movement smoothness, timing consistency, amplitude regularity

Justification:
Beyond tremor, Parkinsonian motor impairment includes deficits in rhythmic movement generation, timing, and coordination (bradykinesia and dysrhythmia). Repetitive motor tasks are widely used in clinical assessments to evaluate these impairments.

Studies have shown that patients exhibit increased variability in movement timing and reduced ability to maintain consistent rhythmic output during cyclic tasks [5]. The rhythmic grip-release task therefore enables assessment of dynamic motor control, complementing the steady-hold task.

The rhythmic grip-release task allows quantification of:

- **Timing accuracy** → deviation from target rhythm
- **Consistency** → variability across cycles
- **Coordination** → smooth transitions between contraction and relaxation

This type of task is particularly sensitive to early motor control deficits and complements steady-hold measurements by evaluating dynamic motor performance.

4 Signal Processing

SIGNAL PROCESSING (SINCE TREMOR IS AT A DIFFERENT FREQUENCY)

4.1 Signal Processing

Accurate feature extraction relies on proper preprocessing of sensor signals to remove noise and highlight physiologically relevant information. Each sensor modality requires a tailored pipeline to ensure reliable measurements.

Pressure Sensor: The force signal is first passed through a low-pass filter (e.g., Butterworth, 10–20 Hz cutoff) to remove high-frequency noise from the sensor and hand vibrations. This ensures that the metrics that we want to extract are not artificially affected by spurious signal fluctuations. Optionally, a baseline correction can be applied to subtract any offset present when the hand is at rest.

EMG: Surface EMG signals are band-pass filtered (typically 20–450 Hz) to eliminate motion artifacts and high-frequency interference. The filtered signal is then rectified to produce a unipolar waveform, which is subsequently smoothed using a moving RMS window (50–250 ms) to generate a clear envelope of muscle activation. Additionally, spectral analyses such as median frequency can be computed to monitor fatigue-related shifts in motor unit recruitment.

IMU: Accelerometer and gyroscope signals are initially low-pass filtered to remove sensor noise. Sensor fusion algorithms, such as quaternion-based orientation estimation, combine accelerometer and gyroscope data to provide accurate estimates of hand orientation and angular velocity while reducing drift. Frequency-domain analyses, including Fast Fourier Transform (FFT) and Power Spectral Density (PSD), are applied to quantify tremor and cyclic movement characteristics. Tremor-related metrics focus on the 4–12 Hz frequency band, which captures clinically relevant oscillations. The main bands are:

- **Physiological tremor: 8–12 Hz** → normal, non-pathological oscillations
- **Parkinson's disease rest tremor: 4–6 Hz** → pathological rest tremor
- **Essential tremor: 5–12 Hz** → high-amplitude postural or action tremor

Voluntary movements and postural adjustments below 4 Hz are ignored, ensuring that only pathological or clinically meaningful tremors contribute to the analysis. Other metrics, such as hold stability, cycle amplitude, smoothness, and hesitation indices, are also derived from the fused orientation and angular velocity signals. s By applying these preprocessing steps, the system ensures that all extracted features—force, EMG activation, and motion—are physiologically meaningful and suitable for clinical interpretation.

5 Feature Extraction

This chapter is devoted to an overview of the main parameters that we can obtain from the signals of each sensor. It should be understood that this overview does not represent a final conclusion regarding the list of parameters that we will use. It is a review of those metrics that seem to us the most informative at the first stage of system design and that will be more useful for building a basic threshold-based algorithm. As the project develops and becomes more advanced, many other metrics will also be investigated in terms of their potential informativeness and usefulness. In addition, a certain part of the metrics will be subjected to filtering during the training of the machine learning model, if required by the process of model optimization and improvement of its efficiency.

5.1 Pressure Feauture

The pressure sensor allows us to analyse how force is produced over time, not only how strong the patient is. From this signal, several meaningful features are extracted to better understand motor performance and rehabilitation progress.

First, the **Maximum Voluntary Contraction (MVC)** represents the peak force achieved during a maximal squeeze. Grip strength is widely considered a global indicator of neuromuscular health and overall functional capacity. In addition to reflecting strength recovery, MVC can also be used to normalize EMG signals, which may vary due to electrode placement or day-to-day physiological differences. Tracking MVC over multiple sessions allows us to monitor rehabilitation: an increase suggests improved strength, while a decrease may indicate deterioration or fatigue [1].

Second, the **Rate of Force Development (RFD)** is calculated as the slope of the force curve:

$$RFD = \frac{dF}{dt} \tag{1}$$

This metric reflects how quickly force can be generated and is closely linked to neural activation speed and motor unit recruitment. Importantly, neural adaptations often occur before visible muscle growth. For this reason, RFD can act as an early biomarker of improvement during rehabilitation, sometimes progressing even before MVC increases.

Third, a **Fatigue Index** is derived from sustained or repeated contractions by measuring the relative decline in force over time. A simple formulation is:

$$\text{Fatigue Index} = \frac{F_{\text{initial}} - F_{\text{final}}}{F_{\text{initial}}} \tag{2}$$

Improved endurance across sessions indicates successful therapy, whereas rapid force decay may suggest neuromuscular impairment or reduced resistance to fatigue [9].

Fourth, **Force Variability** is assessed using measures such as standard deviation (SD), root mean square deviation, and the coefficient of variation:

$$CV = \frac{SD}{\text{Mean Force}} \tag{3}$$

These metrics quantify how steady the force output is during submaximal holds. Higher variability is generally associated with impaired motor control or tremor, while reduced variability reflects improved stability. The coefficient of variation is particularly useful because it accounts for differences in mean force between sessions.

5.2 EMG Features

The EMG channel provides a direct view of muscle activation dynamics and allows us to quantify tremor-related oscillations during gripping. Since raw EMG is broadband and strongly affected by electrode placement and impedance, we first compute an EMG envelope that reflects overall activation over time. In practice, the signal is band-pass filtered, rectified, and low-pass filtered to obtain the

envelope $e(t)$. All tremor-related features are then extracted from the spectrum of this envelope within the tremor band (2–12 Hz).

Tremor Band Power represents how much tremor-related energy is present in muscle activation. It is defined as the integrated power spectral density (PSD) of the envelope within the tremor band:

$$P_{\text{trem}} = \int_{f_1}^{f_2} S_e(f)\, df, \qquad f_1 = 2\,\text{Hz},\ f_2 = 12\,\text{Hz} \tag{4}$$

where $S_e(f)$ is the PSD of the envelope $e(t)$. Higher values indicate stronger tremor modulation.

The Peak Tremor Frequency captures the dominant tremor rhythm by locating the maximum of the envelope spectrum inside the tremor band:

$$f_{\text{peak}} = \arg \max_{f \in [f_1, f_2]} S_e(f) \tag{5}$$

Different tremor mechanisms often manifest with distinct characteristic frequencies.

Tremor Power Ratio - To make tremor strength comparable across subjects and sessions, we normalize tremor power by the total envelope power:

$$R_{\text{trem}} = \frac{\int_{f_1}^{f_2} S_e(f)\, df}{\int_0^{f_{\max}} S_e(f)\, df} \tag{6}$$

This normalization improves robustness to electrode impedance and overall activation differences.

Spectral Entropy in Tremor Band - Spectral entropy measures whether tremor activity is concentrated around a single frequency (periodic) or distributed across many frequencies (noisy). First, a normalized spectral distribution is defined:

$$p_k = \frac{S_e(f_k)}{\sum_{f_i \in [f_1, f_2]} S_e(f_i)} \tag{7}$$

Then Shannon entropy is computed as:

$$H_{\text{trem}} = -\sum_{f_k \in [f_1, f_2]} p_k \log(p_k) \tag{8}$$

Lower entropy corresponds to cleaner and more periodic tremor oscillations.

Tremor Periodicity via Autocorrelation - Tremor rhythmicity can also be captured in the time domain using the autocorrelation of the envelope:

$$r(\tau) = \frac{\sum_t (e(t) - \bar{e})(e(t-\tau) - \bar{e})}{\sum_t (e(t) - \bar{e})^2} \tag{9}$$

We then estimate tremor periodicity as the strongest repeating peak within tremor-related lags:

$$\rho_{\text{trem}} = \max_{\tau \in [\frac{1}{f_2}, \frac{1}{f_1}]} r(\tau) \tag{10}$$

Higher values indicate a more rhythmic and repetitive tremor pattern.

Together, these features provide complementary descriptions of tremor strength, frequency, spectral structure, and temporal periodicity, making them suitable both for threshold-based analysis and machine learning models.

5.3 IMU Features

The IMU sensor provides rich information about hand and wrist motion, complementing pressure and EMG data by quantifying tremor, compensation, and movement quality. From the 3-axis acceleration ($\mathbf{a}_{mag}$) and 3-axis angular velocity ($\mathbf{w}_{mag}$), several features can be extracted to assess motor performance and rehabilitation progress.

Tremor Power (4–12 Hz) is computed as the total spectral energy in the tremor frequency band from the acceleration magnitude:

$$\text{Tremor Power} = \sum_{f=4}^{12} PSD(\mathbf{a}_{mag}) \tag{11}$$

This metric reflects the overall "energy" of involuntary oscillations. Higher tremor power indicates more severe tremor, commonly observed in Parkinson's disease or post-stroke motor instability [4].

Tremor Frequency is extracted as the peak frequency within the 4–12 Hz band:

$$f_{\text{tremor}} = \arg \max_{f \in [4,12]} PSD(\mathbf{a}_{mag}) \tag{12}$$

This identifies the dominant tremor frequency and allows clinical classification: 4–6 Hz for Parkinsonian rest/postural tremor, 5–12 Hz for essential tremor, and outside this range for voluntary or artifact movement.

Orientation Stability is measured as the variance of quaternion-derived angles during a hold phase:

$$\text{Orientation Stability} = Var(\theta_{quaternion}) \tag{13}$$

This quantifies hand drift during static holds. Lower variance indicates better postural control, while higher variance reflects fatigue or compensatory movements. Typical clinical ranges are: $<2^\circ$ for healthy holds, 2–8° for stroke-related drift, and 5–20°+ for Parkinson's postural tremor.

Range of Motion (ROM) is computed as the difference between maximum and minimum angles during maximal grip or dynamic tasks:

$$ROM = \theta_{max} - \theta_{min} \tag{14}$$

Excessive wrist motion can indicate compensation for weak grip strength. Healthy execution typically involves $<5^\circ$, while $>15^\circ$ suggests excessive compensatory movement.

Hold Stability is assessed using the standard deviation of angular velocity during the plateau phase of a hold:

$$\text{Hold Stability} = SD(\mathbf{w}_{mag}) \tag{15}$$

Higher variability indicates tremor or co-contraction during static force output.

Drift Rate quantifies linear trends in hand orientation over time:

$$\text{Drift Rate} = \frac{\theta_{end} - \theta_{start}}{t_{hold}} \tag{16}$$

This captures fatigue-related compensatory movements or loss of postural control during sustained contractions.

Cycle Metrics (for rhythmic tasks) include:

- **Cycle Frequency:** Spectral peak of $\mathbf{w}_{mag}$ in 0.5–2 Hz band, reflecting the speed of repetitive grip-release movements.
- **Cycle Amplitude:** Peak-to-peak angular velocity per detected cycle, indicating hypometria or reduced movement size.
- **Smoothness:** Number of sub-movements per cycle (peaks in $\mathbf{w}_{mag}$ above 10% of max), reflecting jerky or multi-step motion.
- **Hesitation Index:** Fraction of cycle spent above 10% peak velocity, capturing movement sequencing deficits.

These features collectively allow quantification of tremor severity, postural control, bradykinesia, and compensatory strategies. Tracking them over multiple sessions can provide objective feedback on motor recovery and rehabilitation effectiveness.

6 Feedback and Decision Algorithm

A core aspect of the Smart Hand Rehabilitation System is the ability to provide real-time feedback and guide patients during exercises. Feedback serves as a bridge between measured performance and actionable guidance, allowing patients to adjust their movements and effort to optimize rehabilitation outcomes.

In general, feedback and decision algorithms analyze sensor signals to extract relevant features. These features are then compared against predefined thresholds or target values to generate immediate feedback for the patient.

The primary purposes of this approach are:

- To provide patients with interpretable and actionable cues during rehabilitation exercises.
- To objectively quantify performance, progress, and deficits over time.
- To standardize measurements across sessions and between patients.

Depending on the implementation strategy, feedback can be generated using:

- **Rule-Based Algorithms:** Predefined thresholds and logic derived from clinical guidelines and literature.
- **Machine Learning Algorithms:** Adaptive, data-driven models that can personalize guidance based on historical performance and feature patterns.

6.1 Phase 1: Rule-Based System

The first phase of the Smart Hand Rehabilitation System focuses on developing a fully functional platform that accurately captures neuromuscular features and provides immediate, clinically relevant feedback to patients. Phase 1 relies on established thresholds, clinical rules, and feature-based evaluation to guide rehabilitation.

sIn Phase 1, the primary focus is on designing a feedback algorithm capable of interpreting sensor data in real time and providing actionable guidance during rehabilitation exercises. The algorithm takes as input a set of features extracted from the Pressure, EMG, and IMU sensors.

Once the features are computed, the algorithm evaluates them against pre-established clinical thresholds or target values. For instance, the percentage of time the force remains within the target band is compared to an 80% threshold to determine successful execution, while fatigue indices exceeding 15% trigger a recommendation to rest. Tremor amplitude and orientation variance are used to detect instability or compensatory movements, prompting stabilization cues. Cycle amplitude and smoothness metrics identify hesitations or jerky transitions, guiding coordination feedback.

The algorithm then translates these evaluations into both real-time and post-trial feedback. During the exercise, patients receive immediate visual or textual cues, such as force traces overlaid with target bands, and auditory signals that reinforce correct execution or signal deviations. After each trial, a summary is presented quantifying endurance, steadiness, tremor control, and coordination. Importantly, the rules allow for adaptive guidance between trials: for example, achieving a high percentage of time within the target band may result in the algorithm recommending a slightly increased force target in subsequent repetitions.

Clinically, this rule-based approach ensures that feedback is interpretable, actionable, and grounded in evidence-based thresholds. It provides patients with real-time guidance to improve force control, reduce compensatory movements, and enhance endurance, while simultaneously generating a dataset of validated features. This dataset forms the foundation for Phase 2, where machine learning algorithms will leverage these measurements for personalized and autonomous exercise adaptation.

6.2 Phase 2: Machine Learning (Planned)

Machine learning is one of the goals for further deepening and extending this project, since it allows us to build an efficient tremor-detection algorithm that can take into account a large number of diverse parameters. Part of these parameters was described in the previous chapters, however their overall pool will be significantly expanded and later additionally verified and filtered in order to achieve optimization and high model efficiency.

At this moment, there is no final decision regarding which models will be used. A preliminary candidate is SVM, as a compromise between computational complexity and performance, together with LOSO for better learning on a limited dataset. If the features demonstrate good clustering and separability, we may also consider using K-NN.

Regarding the data, it is obvious that within the framework of a university project we do not have access to tremor patients for direct data collection. Nevertheless, we can use public data sources, such as open datasets and scientific articles dedicated to tremor research. For example, we refer to the dataset *"EMG and accelerometer recordings during closed-loop TMS for tremor in Essential tremor and Parkinson's disease"* and to the article *"Quality parameters for a multimodal EEG/EMG/kinematic brain-computer interface (BCI) aiming to suppress neurological tremor inupper limbs"* [10, 6]. Both sources contain very useful tremor measurements based on EMG and IMU sensors, as shown in the figure below, which we can use for training our model. Of course, the data will be split into training and validation parts for a more well-structured model development process.

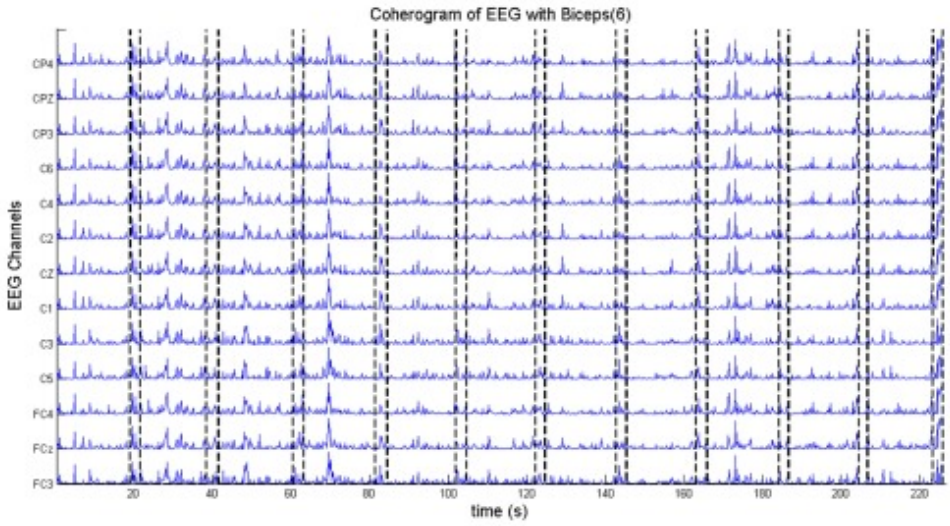


Figure 4: EMG data from the article [6]

7 Web Application

The Smart Hand Rehabilitation System includes a web-based application designed to provide real-time visualization and feedback during rehabilitation exercises. The web interface connects directly to the embedded device via Bluetooth Low Energy (BLE), enabling wireless data transmission without the need for additional software installation. Once connected, the application streams live sensor data—including pressure, accelerometer, and gyroscope measurements—and displays them in an intuitive format for both patients and clinicians.

The primary purpose of the web application is to allow the patients to log into their profiles and observe how they are performing. They can see the values of their biomarkers and if they are improving or deteriorating compared to the previous time they logged in and performed the exercises.

Right now, our webapp looks like the following figure 5:



Figure 5: Current Web-app

The current Web-app, displays the raw values that the sensors are reading. We were using it to fully understand the sensor, verify the wiring and test if our prototype is working as desired.

The idea would be to have patients be able to log into their profiles, where all previous exercises performed are logged. Once they perform the exercises, the website would display all of the new biomarkers values, and using the ML algorithm, will be able to determine if the rehabilitation therapy is indeed working on the patient or not.

The web-app will also display the category the patients stands in since the exercises differ depending on the category of the tremor.

The figure 6 bellow, if an idea of how our future Web-app will look like.

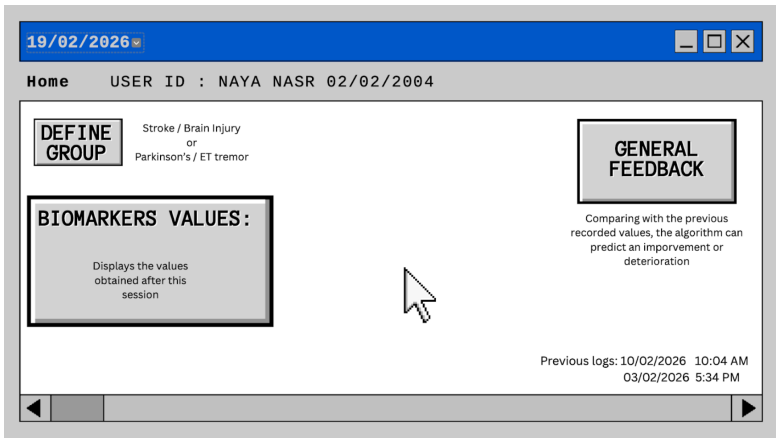


Figure 6: Idea for future Web-app

8 Conclusion

In this project, we presented the design and initial implementation of a Smart Hand Rehabilitation System aimed at improving the assessment and monitoring of patients with neurological disorders such as stroke and Parkinson's disease. Unlike traditional dynamometers that focus only on maximal grip strength, our system integrates pressure, EMG, and IMU sensors into a portable Bluetooth-enabled device capable of capturing both mechanical and neuromuscular aspects of hand function.

We described the full system architecture, including hardware design, sensor integration, power management, and Bluetooth communication between the embedded device and the computer.

We also defined specific experimental protocols tailored to two target populations: stroke/brain injury patients and tremor-dominant patients. For each group, we proposed exercises designed to extract clinically meaningful biomarkers such as Maximum Voluntary Contraction (MVC), Rate of Force Development (RFD), fatigue index, and many more biomarkers.

The web interface serves as both a validation tool for hardware testing and a foundation for future patient-oriented dashboards, where users will be able to log in, track their biomarkers, and monitor improvement across sessions.

However, this project is still in development. The next major step is the implementation of machine learning algorithms to move beyond fixed thresholds and toward adaptive, data-driven decision-making. By training models, we aim to build a more robust classification system capable of detecting subtle improvements that may not be visible through simple threshold analysis. Feature selection and model optimization will play a crucial role in ensuring efficiency and generalizability.

Our ultimate goal is to provide patients and clinicians with an accessible, objective, and data-driven system that supports rehabilitation, tracks recovery more precisely, and contributes to better therapeutic outcomes.

While further testing, validation, and refinement are required, the current results demonstrate the feasibility of combining multimodal sensing, signal processing, and intelligent feedback into a compact and scalable rehabilitation platform.

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